DUALFORCE Hybrid Engine: Materials & Parts Catalog

Complete Technical Specification of Parts, Advanced Materials, Engineering Functions & Physical Advantages

Powertrain Architecture: Single-Chamber Combustion + Linear Reluctance
Displacement Range: 97.2 cc (Commuter) to 648 cc (Supersport)
Document Type: Engineering Parts & Material Science Specification
Target Audience: Automotive R&D, Faculty Review & Manufacturing Tenders

EXECUTIVE MATERIALS SUMMARY The DUALFORCE Hybrid Engine achieves complete unification of internal combustion and electromagnetic reluctance without auxiliary traction motors. This is made possible through deliberate material science pairings: **NITRONIC 50 NON-MAGNETIC STEEL** prevents flux shunting; **45 MM NI-SIC NIKASIL MATRIX** eliminates stainless steel galling; **LAMINATED HIPERCO-50** delivers 2.4 T saturation with a 940°C Curie threshold; **YSZ CERAMIC PLASMA-SPRAYING** reflects 68% of flash combustion heat; and **MIDEL 7131 SYNTHETIC DIELECTRIC ESTER** ensures fire-safe, non-conductive liquid cooling.

1. CYLINDER BORE & ELECTROMAGNETIC STATOR ASSEMBLY

Component Part	Material Formulation	Engineering Usage / Function	Physical & Performance Advantage
Cylinder Liner / Sleeve	**NITRONIC 50 (XM-19)** or Axially-Slotted Austenitic 316L	Forms the rigid cylinder bore wall separating the combustion chamber from the surrounding stator coils.	• **Non-Magnetic:** μr ≈ 1.004–1.006 allows flux to pass straight into the piston with zero shunting. • **Zero Martensite:** High nitrogen suppresses strain-induced martensite (α') during plateau honing. • **Eddy-Current Slits:**

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			Axial dielectric micro-slits suppress circumferential eddy currents by >90%.
Bore Running Surface	**45 MM NI-SIC (NIKASIL)** Composite Matrix Coating	Direct contact friction surface for piston rings and skirt within the stainless liner sleeve.	• **Hardness >65 HRC:** Permanently eliminates the severe adhesive galling notorious in raw stainless steel. • **Oil Retention:** Oleophilic silicon-carbide micropores sustain a continuous hydrodynamic oil film ($\mu \approx 0.03$).
Stator Drive Windings	**CLASS-220 TYPE-2** Copper Litz Wire	Multi-tier circumferential coils wound outside the liner that generate bidirectional reluctance thrust and regenerative braking.	• **Eliminates AC Skin Effect:** Individually insulated micro-strands cut high-frequency resistance above 6,000 RPM. • **Thermal Margin:** Rated to 220°C (operates at 114°C steady, leaving a +106°C safety headroom).
Stator Back-Iron Core	**0.20–0.35 MM M250-35A** Silicon Steel Laminations (C-5)	Encircles copper coils to complete the magnetic flux return path outside the cylinder.	• **Low Core Loss:** Thin laminations cut eddy-current iron losses. • **Flux Directivity:** Focuses magnetic lines radially inward through the liner into the piston

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			teeth with minimal leakage flux.
Stator Cooling Bores	**MIDEL 7131** Synthetic Dielectric Ester	Circulates non-conductive dielectric fluid at 16 L/min through 24 vertical laser-drilled jacket bores.	• **Electrically Non-Conductive:** Breakdown strength >75 kV eliminates short circuits. • **Fire Safe:** High fire point (>310°C). • **Direct Cooling:** Directly cools stator copper to 108–114°C under maximum dyno load.

2. RECIPROCATING ASSEMBLY (PISTON & RELUCTANCE ARMATURE)

Component Part	Material Formulation	Engineering Usage / Function	Physical & Performance Advantage
Piston Body & Crown	**FORGED 4032 AEROSPACE** Aluminum Alloy	Primary reciprocating structural body absorbing 75 bar combustion pressure and housing the reluctance teeth.	• **High Silicon Content (12%):** Low thermal expansion (19.4 × 10^{-6}/K), high fatigue strength, and high scuff resistance. • **Non-Magnetic:** Paramagnetic structure prevents unbalancing magnetic drag.

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Crown Thermal Barrier (TBC)	**0.4 MM YSZ** Yttria-Stabilized Zirconia	Plasma-sprayed onto the top combustion-facing surface of the piston crown.	• **Low Thermal Conductivity (k ≈ 1.8 W/m·K):** Reflects up to 68% of radiant combustion flash heat back into gas expansion. • **Insulates Reluctance Core:** Protects underlying aluminum and reluctance teeth from 2,000°C flame.
Reluctance Translator Teeth	**0.2 MM LAMINATED HIPERCO-50** (49% Fe, 49% Co, 2% V)	Ferromagnetic teeth embedded along the piston skirt that get pulled axially by the stator coils.	• **Highest Saturation (2.4 Tesla):** Maximizes linear force per unit area (0.573 MPa magnetic pressure). • **Zero Residual Magnetism:** Coercivity Hc < 12 A/m ensures 0.00 mT remanence at cut-off (zero magnetic clamping). • **940°C Curie Threshold:** Completely immune to thermal demagnetization (unlike NdFeB magnets that die at 150°C).
CTE Expansion Joint	**AXIAL DOVETAIL TRACKS** + Inconel 718 Belleville Clips	Mechanically anchors the Hiperco-50 laminations onto the aluminum piston skirt.	• **Compensates Thermal Mismatch:** Aluminum expands at 19.4×10^{-6}/K vs. Hiperco at 9.5×10^{-6}/K. • **Zero Shear Stress:** Floating dovetail allows

Component Part	Material Formulation	Engineering Usage / Function	Physical & Performance Advantage
			skirt to expand by 38 μm at 200°C without shearing or ovalizing.
Piston Skirt Coating	**DIAMOND-LAPPED MOS_2 / PTFE** Polymer Solid Matrix	Applied to the outer contact thrust faces of the piston skirt.	• **Dry Lubricity (μ = 0.03):** Eliminates scuffing during cold starts or transient lateral tilt. • **Fluid Inertness:** 100% chemically inert to both motor oil and dielectric ester.
Piston Ring Pack	**HIGH-ALLOY STEEL + PVD CRN** Low-Tension Racing Pack	Top compression and oil control rings sealing combustion gases.	• **Low Radial Tension (<0.14 MPa):** Cuts parasitic ring friction drag by 38% during pure-EV city mode. • **CrN PVD Hardness (>2,000 HV):** Exceptional wear resistance against the Nikasil-plated liner.
Wrist Pin (Gudgeon Pin)	**TITANIUM 6AL-4V** + Diamond-Like Carbon (DLC)	Transfers reciprocating piston force into the connecting rod small-end.	• **Lightweight (32 g):** 45% weight saving over traditional steel pins. • **DLC Low Friction:** Minimizes pin-boss boundary friction under extreme combustion shock.

3. VALVETRAIN, GAS EXCHANGE & CYLINDER HEAD

Component Part	Material Formulation	Engineering Usage / Function	Physical & Performance Advantage
Dynamic Camshaft	**BILLET 8620 CARBURIZED STEEL** Dual-Lobe VVA Profile	Controls intake/exhaust valve timing, Atkinson over-expansion, and EV mode decompression.	• **Atkinson/Miller Over-Expansion:** Delays IVC to 48° ABDC, extracting +18.4% more thermodynamic work. • **Zero-Lift Decompression:** Decouples rocker arms via pin-latches in EV mode, slashing pumping drag to <180 W.
Intake & Exhaust Valves	**TI-6AL-4V (INTAKE)** Inconel 751 (Exhaust)	Poppet valves regulating air-fuel induction and high-velocity exhaust expulsion.	• **40% Lighter Valvetrain:** Titanium intake valves prevent valve float at 10,500 RPM redline. • **Hot Strength:** Inconel 751 exhaust valves resist continuous 850°C corrosive exhaust gases without recession.
Cylinder Head & Jacket	**CAST A356-T6 ALUMINUM** + 50/50 Water-Ethylene Glycol	Forms combustion chamber roof, spark plug boss, and valve ports.	• **High Heat Capacity (cp ≈ 3.6 kJ/kg·K):** Water-glycol loop absorbs 65% of total combustion heat rejection, holding head temperatures steady at 88–94°C.

4. KINEMATICS, CRANKCASE & POWER TRANSMISSION

Component Part	Material Formulation	Engineering Usage / Function	Physical & Performance Advantage
Crankshaft	**FORGED 4340 CHROME-MOLY** Steel (270° Crossplane)	Converts linear piston thrust into rotational mechanical shaft power.	• **High Fatigue Strength (>1,100 MPa):** Easily withstands combined 75 bar combustion + 3,680 N reluctance pulses. • **Phase-Shifted Torque:** Reluctance pulse timed at 35°–75° ATDC smooths cyclic torque ripple by 42%.
Connecting Rods & Bearings	**FORGED H-BEAM 4340** + Caged Needle-Roller Bearings	Mechanical linkage between wrist pin and crankshaft journal throws.	• **Reduced Friction:** Caged needle-roller bearings eliminate hydrodynamic sleeve drag during low-RPM EV city crawl. • **Buckling Resistance:** H-beam profile withstands bidirectional alternating loads.
Balance Shaft	**HARDENED CARBON STEEL** Gear-Driven Counter-Rotator	Positioned in the lower front crankcase driven off primary crank gear.	• **Inertial Balancing:** Completely cancels the primary rocking couple and transverse shaking forces inherent to 270° parallel twins.

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Engine Block / Crankcase	**HIGH-PRESSURE DIE-CAST /** CNC 6061-T6 Aircraft Aluminum	Outer skeleton housing the crankcase, modular stator cartridge, and 6-speed transmission.	• **Lightweight Backbone (<18 kg):** High rigidity-to-weight ratio. • **Faraday EMI Shield:** Aluminum housing acts as an RF enclosure, shielding ECU and CAN bus from high-frequency inverter switching noise.

5. CRITICAL MATERIAL PROPERTIES REFERENCE MATRIX

Engine Material	Relative Permeability (μr)	Thermal Conductivity (k)	Critical Threshold / Hardness
Nitronic 50 (Cylinder Sleeve)	μr ≈ 1.004 – 1.006	14.0 W/m·K	Md30 < -50°C (No Martensite)
Ni-SiC Nikasil (Bore Coating)	Non-Magnetic (μr = 1.00)	~60.0 W/m·K	Hardness >65 HRC (>800 HV)
Hiperco-50 (Reluctance Teeth)	μr(max) ≈ 10,000 (Bsat = 2.4T)	29.8 W/m·K	Curie Temp = 940°C (1,213 K)

Engine Material	Relative Permeability (μr)	Thermal Conductivity (k)	Critical Threshold / Hardness
YSZ Ceramic (Piston Crown)	Non-Magnetic (μr = 1.00)	1.80 W/m·K	Sintering Limit = 1,200°C
Forged 4032 (Piston Body)	Paramagnetic (μr ≈ 1.000)	154.0 W/m·K	Yield Strength = 315 MPa @ 150°C
MIDEL 7131 (Dielectric Coolant)	Non-Magnetic (μr = 1.00)	0.144 W/m·K	Breakdown Voltage >75 kV

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